

Three voyage PDFs in. One defensible demurrage claim out. Under a minute.

An AI agent that resolves maritime laytime and demurrage, built for Greece, the world's largest shipowning nation.



AI-first claims platform for maritime operations
Built by two engineers at the ACG AI Lab, Athens

LIVE DEMO CLAIM, REPRODUCIBLE

EUR 84,375.00

THE PAIN

Demurrage costs tens of thousands a day per ship, still resolved by hand in Excel and email.

When a port call ends and laytime is disputed, a claims team spends two to four days assembling the Charter Party, Notice of Readiness, and Statement of Facts by hand. Laytime is one of the most frequently disputed areas of maritime law.

\$30–80k³

Demurrage per idle ship per day, Capesize to VLCC.

90 days²

Typical time bar. Miss it, the claim is forfeited.

2–4 days¹³

Analyst time per disputed claim, today, by hand.

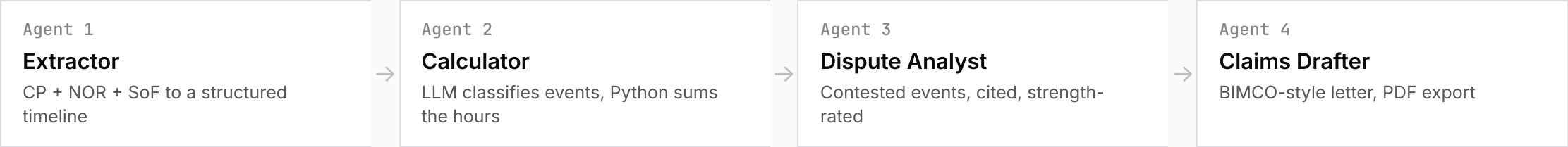
~20%¹

Of the world fleet is Greek-owned (61% of EU).

Cite: [1] UGS Annual Report 2024–25 / Petrofin · [2] ITIC & Watson Farley & Williams · [3] Voyager Portal (day rates) · [13] Laytimely (status quo we observe). Full URLs on the sources slide.

WHAT RUNS TODAY LIVE

A four-agent pipeline turns three PDFs into a citation-backed claim packet, and the arithmetic is auditable, not hallucinated.



NOR tendered, laytime counting crossover → EUR 1,875 / h

Auditable	Hours summed in deterministic Python, locked by a test. Not LLM math.
Cited	Every line references its CP clause number and SoF event id.
Time bar	The 90-day deadline surfaced on every claim.

DEMO VOYAGE · MT AEGEAN PIONEER · RAS TANURA → ROTTERDAM

EUR 84,375.00

Reproducible: feeding the three demo PDFs through the live pipeline returns stage=done at exactly this figure. INTERNAL, LIVE ¹³

The leap: from a tool you upload to, to an autonomous claims co-worker, inbox to outbox.

WHY WE WIN

Incumbents bolted AI onto forms.
We are agent-native, with
auditable math none of them claim.

PLAYER	RAISED / SCALE	POSITION, AND THE GAP WE EXPLOIT
Veson Nautical	~\$102M rev ⁹	The IMOS gold standard (Francisco Partners + Pamlico), but enterprise-only and its AI is a bolt-on. Leaves the SME owner long tail open.
Burmester & Vogel MOST DIRECT	private ¹²	A "Demurrage AI Copilot," consolidating the niche (acquired Laysoft, 2025). A copilot that assists, not an autonomous, sub-minute packet.
Voyager Portal	~\$12.7M ¹¹	Charterer-focused. Its AI is an image-recognition SoF parser, not end-to-end reasoning, dispute, and drafting.
Harbor Lab SIBLING	~\$22M ¹⁰	Athens, Atomico and TMV backed, validates port costs . We resolve demurrage . Adjacent workflow, natural partner, not a rival.

OUR MOAT	DEFENSIBLE	ENDGAME
Agent-native pipeline plus deterministic, auditable arithmetic that an LLM-only tool cannot match on trust.	Citation-backed legal output, time-bar engine, EU-first and EUR-native from day one.	A two-sided settlement network: owners and charterers resolving claims on one platform.

Cite: [9] getLatka / PitchBook / Francisco Partners · [10] TechCrunch / Atomico · [11] GlobeNewswire / PitchBook · [12] Smart Maritime Network. Private-company revenues are third-party estimates.

UNIT ECONOMICS

Each claim costs us about EUR 0.07 to produce and sells inside a ~90% gross-margin model.

REVENUE LINE	PRICE
SaaS workspace	EUR 299/mo
Success fee on recovered demurrage	5%
Claims Radar, live exposure	EUR 299/vessel/mo
EU ETS / FuelEU compliance	EUR 50-100/voyage
A mid-size owner (15 vessels)	EUR 50-130k ARR

Land on SaaS plus success fee, expand seat to fleet with Radar, then compliance. Net revenue retention above 120%. **ILLUSTRATIVE**

WHY IT COMPOUNDS

EUR 0.07	Cost to produce one claim: Claude plus prompt caching. ¹³
~90%	Gross margin. Marginal cost is effectively zero.
Owners	We sell first, in Greece. Revenue recovery is a money-in pitch.
Then	Charterers, maritime lawyers, the two-sided network.

One recovered mid-size claim, tens of thousands of euros, pays for a year of Laytimely. Beyond fixed costs, nearly every new euro is profit.

Cite: [13] Laytimely live pipeline (internal). Pricing and ACV are an illustrative model for discussion.

TAM · SAM · SOM

A Greece-first wedge in maritime commercial software, with a regulated compliance market opening next door.

TAM · Maritime commercial software + EU compliance

\$0.3–4.7B + €10B/yr^{5,6,7}

Voyage-management software is ~\$0.3–2.3B today at 7–14% CAGR (analysts disagree on base size); broader maritime software reaches ~\$4.7B by 2033. Next door, EU ETS adds up to **€10B/yr** of cost to shipping from 2026, and FuelEU penalties €2,400/t⁸: a compliance market on the same pipeline.

SAM · EU owners, charterers & maritime lawyers

~EUR 100–150M ARR

Voyage-charter operators in the EU-controlled fleet; Greece alone is 61% of it¹.
Bottom-up: addressable orgs × blended ACV. **ILLUSTRATIVE**

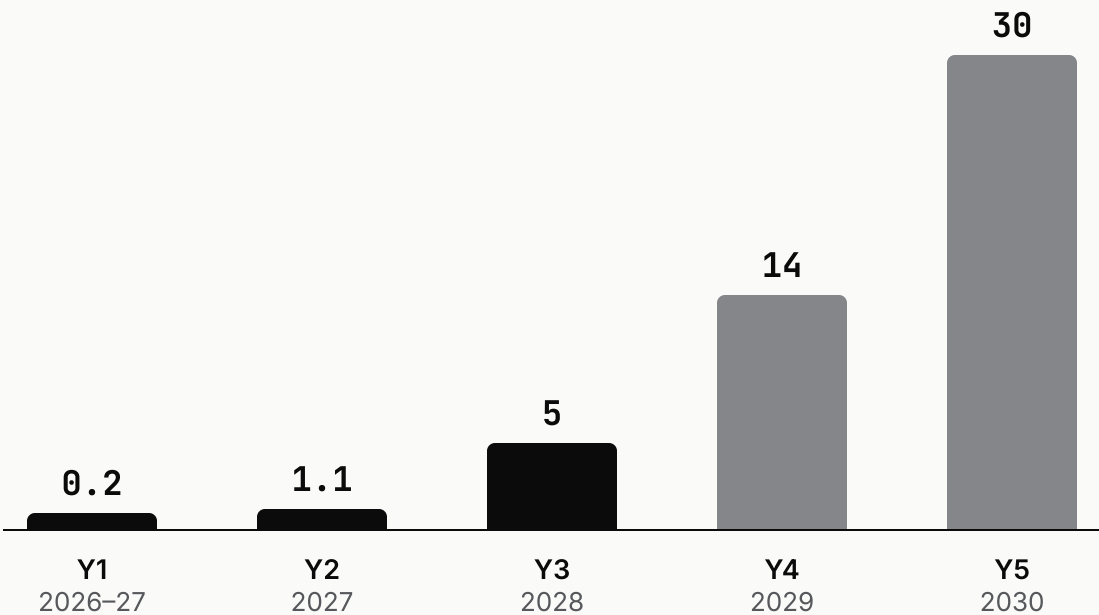
SOM · Obtainable in 5 years

~EUR 30M ARR

A few hundred customers across Greece and the EU, before global charterer expansion. **ILLUSTRATIVE**

FIVE-YEAR MODEL **ILLUSTRATIVE, ASSUMPTIONS STATED**

From design partners to ~EUR 5M ARR by year 3 and ~EUR 30M by year 5.



ARR in EUR millions.

YEAR	CUSTOMERS	ARR	EBITDA
Y1	5	0.2	(0.5)
Y2	25	1.1	(1.0)
Y3	90	5	~0
Y4	200	14	2.5
Y5	360	30	8

Y1 is paid design partners. Assumptions: ~90% gross margin, NRR above 120% via Radar and compliance upsell, low logo churn because the time bar and recovery are mission-critical. EUR millions.

WHAT WE WATCH, AND WHAT WE DO ABOUT IT

Our three real risks are incumbent reaction, data dependencies, and legal defensibility.

Incumbent catches up

Veson or Voyager ships agent-native

A 12 to 18 month window. We lock lighthouse Greek owners and stand up the two-sided network before they move.

Data dependencies

AIS and port feeds for Claims Radar are paid and rate-limited

Start with free weather (Open-Meteo) plus one paid AIS provider; cap voyages per tier in early plans.

Legal defensibility

A wrong number erodes trust fast

Deterministic, cited, auditable output, plus a server-side gate that rejects any letter edit that changes a figure or drops a clause reference.

Single-workflow concentration

Demurrage alone is one line

Compliance and adjacent document workflows reuse the same pipeline: additive schema fields, no second codebase.

BUILT AT THE ACG AI LAB, ATHENS

Two AI researchers whose conviction, verifiable AI that shows its work, is exactly what legal-grade claims demand.

**Dimitris Kalligaridis**

SOFTWARE ENGINEER, AMAZON · AI RESEARCH, ACG AI LAB

Distributed systems, optimization, and applied ML. Built backend infrastructure, authorization, and production performance at Amazon; a Google Summer of Code 2026 contributor to the Eclipse Foundation. Researches reinforcement-learning agents that decide under uncertainty. Make the math explicit and let every number trace back to its source: that conviction is Laytimely's engine.

**Roman Dolgopolyi**

EXPLAINABLE AI & CYBERSECURITY, ACG AI LAB · INCOMING PHD, NORTHEASTERN

Published work on explaining AI decisions in cancer-cell diagnosis, and benchmarks for AI safety compliance. One conviction carries straight into Laytimely: AI trusted with high-stakes decisions has to be transparent and verifiable, never a black box.

Hiring next: a maritime domain anchor (former demurrage analyst or chartering operator). With TMV's ship-owner network, an advisor here is within reach and closes our one real gap.

DISTRIBUTION

We acquire owners through Piraeus relationships and prove ROI with a single recovered claim.

THE MOTION

Reach	Warm intros via investors (TMV, the Hadjipateras owner network), the Union of Greek Shipowners, Posidonia.
Land	Free pilot on a handful of past disputed voyages; show recovered euros.
Convert	To SaaS plus 5% success fee.
Expand	Seat to fleet with Radar, then compliance. NRR above 120%.

WHY GREECE FIRST

The densest concentration of shipowners on earth, ~20% of the world fleet¹, mostly within a few square kilometers of Piraeus. The buyer is physically reachable and relationship-driven, which collapses customer acquisition cost. The long tail of small and mid owners still runs Excel, email, and outside consultants.

Then the same product, charterer perspective already supported, opens the rest of the EU and global commodity traders.

Cite: [1] UGS Annual Report 2024–25 / Petrofin Research.

BUILT IN-HOUSE

A frozen data contract, deterministic math, and prompt caching let us produce claims at cents each.

STACK

Backend	FastAPI, async Python 3.12, Pydantic v2 frozen VoyageState contract, pdfplumber.
Model	Claude Sonnet 4.6 by default, Opus escape per agent. Prompt caching keeps cost at EUR 0.05–0.10 a voyage. ¹³
Frontend	Next.js 15, client-side PDF export, no native deps.
Scale	Stateless agents, Postgres-ready, S3-ready, public API plus webhooks so TMS systems push docs in and pull packets out.

Cite: [13] Laytimely live pipeline (internal).

THE TRUST ARCHITECTURE

The model classifies each event. A plain Python function walks the timeline, sums the hours, splits the row where the allowance runs out, and multiplies by the rate. The number on screen is reproducible and locked by a test to **EUR 84,375.00**.

That separation, LLM for judgment, Python for money, is why a shipping professional can defend our output in a dispute. No competitor that does the math inside the model can say the same.

WHAT IS ALREADY DONE

In one hackathon we shipped a working, tested, multi-tenant product with a reproducible live claim.

PRODUCT

Four-agent pipeline, deterministic calculator with a passing test gate.

Excel, Word, and PDF export; claim-letter email send.

Evidence checklist; claim-strength sub-scores; legal-citation subsystem.

PLATFORM

Multi-tenant workspaces, roles, invitations, audit log.

Email-in ingestion; Postgres-ready persistence; S3-ready storage.

Owner and charterer perspectives both supported.

PROOF

Live demo returns EUR 84,375.00 every run.

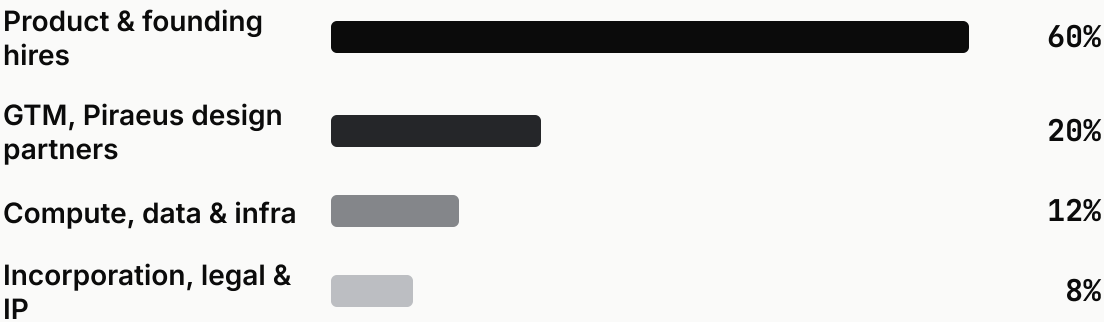
Built at Florent x Panathēnea, Athens, 28 May 2026.

Next: incorporation, IP assignment, and first design-partner pilots.

THE ASK · PRE-SEED

Raising EUR 750k to incorporate, land our first paying design partners, and complete the demurrage product.

USE OF PROCEEDS



Instrument SAFE, target ~EUR 5–6M post-money cap.

Runway 18 months.

WHAT THIS BUYS (TO UNLOCK THE SEED)

Company	Incorporated, IP assigned, first two hires.
Revenue	5+ paying design partners, first ~EUR 250k ARR.
Product	Demurrage complete (multi-port, time charters, OCR); Radar in pilot.
Proof	1 to 2 lighthouse Greek owner references.

ILLUSTRATIVE Terms and milestones for discussion with the lead.

THE CALL TO ACTION

Now: LLMs can finally read legal documents, the EU just mandated emissions reporting, and incumbents are still shipping forms.

TECHNOLOGY

Legal-grade document reasoning became possible only in the last 18 months. The window for agent-native maritime is open now.

REGULATION

EU ETS for shipping (Jan 2024) and FuelEU Maritime (Jan 2025) created an urgent compliance market on the same pipeline.

CAPITAL & POLICY

Maritime is now a declared strategic priority, the Greek fleet is at record size, and the agent-native category is still unclaimed.

WHERE THIS GOES

From one disputed voyage to the operating layer for maritime claims and compliance.

Start with demurrage recovery for Greek owners. Expand to charterers, lawyers, live exposure prevention, EU emissions compliance, then cargo damage and freight disputes, all on one agent pipeline. End with the two-sided settlement network where both parties resolve on Laytimely. That is the moat, and the venture-scale outcome.

EVERY CITED FIGURE, AND WHERE IT COMES FROM

What is sourced, and what is our own model.

INTERNAL

 Laytimely live product, verifiable in the repo

ILLUSTRATIVE

 Our model, assumptions on the slide All others are external sources below

1 Greek fleet ~5,500–5,800 ships, ~20% of world DWT, 61% of EU fleet, ~588 companies.

PRIMARY

Union of Greek Shipowners Annual Report 2024–25 · Petrofin Research

2 Demurrage time bar typically 90 days; claim forfeited if missed.

PRIMARY

itic-insure.com · wfw.com/articles/demurrage-time-bar-clauses

3 Demurrage day rates: Capesize >\$30k, VLCC ~\$50–80k / day.

voyagerportal.com (secondary, vendor explainer)

4 Global seaborne trade ~12.3bn tonnes (2023).

PRIMARY

UNCTAD Review of Maritime Transport 2024

5 Voyage-management software ~\$0.3–2.3B, 7–14% CAGR (range).

Fortune Business Insights · market.us · MarketResearchFuture

6 Maritime software ~\$1.9B (2024) to ~\$4.7B (2033).

globalgrowthinsights.com/market-reports/maritime-software-market

7 EU ETS up to ~€10B/yr cost to shipping (2026); ships ≥5,000 GT; 40/70/100% phase-in.

PRIMARY

DNV · climate.ec.europa.eu

8 FuelEU Maritime penalty €2,400 / t energy deficit (Jan 2025).

PRIMARY

EU Regulation 2023/1805

9 Veson ~\$102M rev / ~\$306M val (est.); Francisco Partners + Pamlico.

getlatka.com · pitchbook · franciscopartners.com

10 Harbor Lab \$16M Series A led by Atomico (~\$22.5M total); TMV investor.

PRIMARY

techcrunch.com/2024/05/15 · atomico.com

11 Voyager Portal \$8.4M Series A (2021), ~\$12.7M total.

globenewswire.com (2021) · pitchbook

12 Burmester & Vogel "Demurrage AI Copilot"; acquired Laysoft (2025).

smartmaritimenetwork.com/2025/03/28

13 EUR 0.05–0.10 / claim; live demo EUR 84,375.00; 2–4 day status quo.

INTERNAL

Laytimely repo: calculator test, README

Deliberately not cited: the widely-quoted "\$22B demurrage" is container detention & demurrage (liner trade), a different, adjacent market. Demurrage-as-%-of-freight and the voyage-charter demurrage \$ pool have no authoritative public source, so we model them bottom-up rather than quote a headline number.